

Embedded and IOT Systems



E&ICT Academy, IIT Kanpur
A Joint Initiative of MeitY & IIT Kanpur

Today's Agenda

- dashboard introduction
- Connection with mqtt broker

Select 2 **mqtt in** nodes and drag them into workspace

The screenshot shows the Node-RED web interface in a browser. The address bar displays the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface includes a top navigation bar with a 'Deploy' button. On the left, a 'filter nodes' sidebar shows a list of nodes, with the 'network' category expanded to reveal 'mqtt in', 'mqtt out', 'http in', 'http response', 'http request', 'websocket in', 'websocket out', and 'tcp in'. The central workspace, titled 'Flow 4', contains two 'mqtt' nodes. The first node is labeled with a green '1' and the second with a green '2'. A green text annotation in the workspace reads 'you need 2 mqtt in for 2 different topics'. On the right, an 'info' sidebar provides details about the current flow, including its ID, name, status, and a description section.

Node-RED

filter nodes

Flow 2 Flow 1 Flow 4

mqtt

mqtt

you need 2 mqtt in for 2 different topics

mqtt in

mqtt out

http in

http response

http request

websocket in

websocket out

tcp in

info

Information

Flow	"aee7e802.c60bb8"
Name	Flow 4
Status	Enabled

Description

Double click on first **mqtt in** node

The screenshot shows the Node-RED web interface in a browser. The address bar displays the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface includes a top navigation bar with a 'Deploy' button. The main workspace shows a flow editor with two tabs: 'Flow 2' and 'Flow 4'. The 'Flow 4' tab is active, and it contains two MQTT nodes. The first MQTT node is highlighted with a red rectangle. The left sidebar shows a list of nodes, including 'trigger', 'exec', 'random', 'rbe', and a 'network' category containing 'mqtt in', 'mqtt out', 'http in', 'http response', 'http request', 'websocket in', 'websocket out', and 'tcp in'. The right sidebar shows the 'info' panel with the following information:

Information	
Flow	"aee7e802.c60bb8"
Name	Flow 4
Status	Enabled

Below the information panel is a 'Description' section, which is currently empty.

Add new mqtt broker(server) here

The screenshot shows the Node-RED web interface at 127.0.0.1:1880. The 'Edit mqtt in node' panel is open, displaying the following configuration:

- Server:** local broker (dropdown menu is open, showing 'local broker', 'test.mosquitto.org:1883', and 'Add new mqtt-broker...' which is highlighted with a red box).
- Topic:** local broker
- QoS:** (empty dropdown)
- Output:** auto-detect (string or buffer)
- Name:** Name

The right sidebar shows the node's information and help:

- Information:** Node ID is "cfcf4892.02dc68", Type is mqtt in.
- Description:** Connects to a MQTT broker and subscribes to messages from the specified topic.
- Node Help:** Connects to a MQTT broker and subscribes to messages from the specified topic.
- Outputs:** payload (string | buffer), topic (string), qos (number), retain (boolean).

Now click on edit button mentioned in slide

The screenshot shows the Node-RED web interface. On the left, the 'network' tab is selected, showing various nodes including 'mqtt in'. In the center workspace, two 'mqtt' nodes are visible. On the right, the 'Edit mqtt in node' configuration panel is open. The 'Server' dropdown menu is expanded, and the 'Add new mqtt-broker...' button is highlighted with a red box. The 'Done' button is also highlighted with a red box. The 'Properties' section shows the following configuration:

- Server: Add new mqtt-broker...
- Topic: Topic
- QoS: 2
- Output: auto-detect (string or buffer)
- Name: Name

The 'Info' panel on the far right shows the node's information:

- Node: "cfcf4892.02dc68"
- Type: mqtt in
- Description: Connects to a MQTT broker and subscribes to messages from the specified topic.
- Node Help: Connects to a MQTT broker and subscribes to messages from the specified topic.
- Outputs: payload (string | buffer), topic (string), qos (number), retain (boolean).

Fill the details

Node-RED interface showing the configuration of a new mqtt-broker node. The configuration details are as follows:

- Name:** uniconverge mqtt (1)
- Server:** test.uniconvergetech.in (2)
- Port:** 1883 (3)
- Client ID:** jitesh (4)
- Keep alive time (s):** 60
- Use clean session:** ☒
- Use legacy MQTT 3.1 support:** ☐

The **Add** button is highlighted in red (5).

Right sidebar information:

- Node:** "a7a1909c.43521"
- Type:** mqtt-broker
- Description:** Configuration for a connection to an MQTT broker. This configuration will create a single connection to the broker which can then be reused by MQTT In and MQTT Out nodes. The node will generate a random Client ID if one is not set and the node is configured to use a Clean Session connection. If a Client ID is set, it must be unique to the broker you are connecting to.
- Birth Message:** This is a message that will be published on the configured topic whenever the connection is established.

Fill the details and select topic **sensor/temperature**

The screenshot displays the Node-RED web interface. In the center, the 'Edit mqtt in node' configuration panel is open. The 'Server' field is set to 'uniconverge mqtt', the 'Topic' field is set to 'sensor/temperature', and the 'QoS' field is set to '0'. The 'Done' button is highlighted with a red box. The right sidebar shows the node's information and help text.

Properties

- Server: uniconverge mqtt
- Topic: sensor/temperature
- QoS: 0
- Output: auto-detect (string or buffer)
- Name: Name

Information

- Node: "cfcf4892.02dc68"
- Type: mqtt in

Description

Connects to a MQTT broker and subscribes to messages from the specified topic.

Node Help

payload *string | buffer*
a string unless detected as a binary buffer.

topic *string*
the MQTT topic, uses / as a hierarchy separator.

qos *number*
0, fire and forget - 1, at least once - 2, once and once only.

retain *boolean*
true indicates the message was retained.

Double click 2nd mqtt in node

The screenshot shows the Node-RED web interface in a browser. The address bar indicates the URL is `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface displays a workspace with two flows: 'Flow 2' and 'Flow 4'. In 'Flow 4', there is a 'sensor/temperature' node and an 'mqtt' node. The 'mqtt' node is highlighted with a red rectangle. The left sidebar shows a list of nodes, including 'trigger', 'exec', 'random', 'rbe', and a 'network' section with 'mqtt in', 'mqtt out', 'http in', 'http response', 'http request', 'websocket in', and 'websocket out'. The right sidebar shows the 'info' panel for the selected 'mqtt in' node. The 'Information' section shows the Node ID as `"1d26ad5a.7bb833"` and the Type as `mqtt in`. The 'Description' section states: 'Connects to a MQTT broker and subscribes to messages from the specified topic.' The 'Node Help' section provides details about the node's inputs and outputs.

Information	
Node	"1d26ad5a.7bb833"
Type	mqtt in

show more

Description

Node Help

Connects to a MQTT broker and subscribes to messages from the specified topic.

Outputs

- payload** (string | buffer): a string unless detected as a binary buffer.
- topic** (string): the MQTT topic, uses / as a hierarchy separator.
- qos** (number): 0, fire and forget - 1, at least once - 2, once and once only.
- retain** (boolean): true indicates the message was retained.

Select **uniconverge mqtt** as a server

The screenshot shows the Node-RED web interface. In the center, the 'Edit mqtt in node' dialog is open. The 'Server' dropdown menu is expanded, and 'uniconverge mqtt' is highlighted with a red rectangle. Other options in the dropdown include 'local broker', 'test mosquitto.org:1883', and 'Add new mqtt-broker...'. The 'Topic' field is set to 'local broker', 'QoS' is set to 'auto-detect (string or buffer)', and the 'Name' field is empty. The 'Enabled' checkbox is checked.

Properties

- Server: local broker (selected)
- Topic: local broker
- QoS: auto-detect (string or buffer)
- Name: Name

☒ Enabled

Information

Node	"1d26ad5a.7bb833"
Type	mqtt in

Description

Connects to a MQTT broker and subscribes to messages from the specified topic.

Outputs

- payload** (string | buffer): a string unless detected as a binary buffer.
- topic** (string): the MQTT topic, uses / as a hierarchy separator.
- qos** (number): 0, fire and forget - 1, at least once - 2, once and once only.
- retain** (boolean): true indicates the message was retained.

Fill the details and select topic **sensor/humidity**

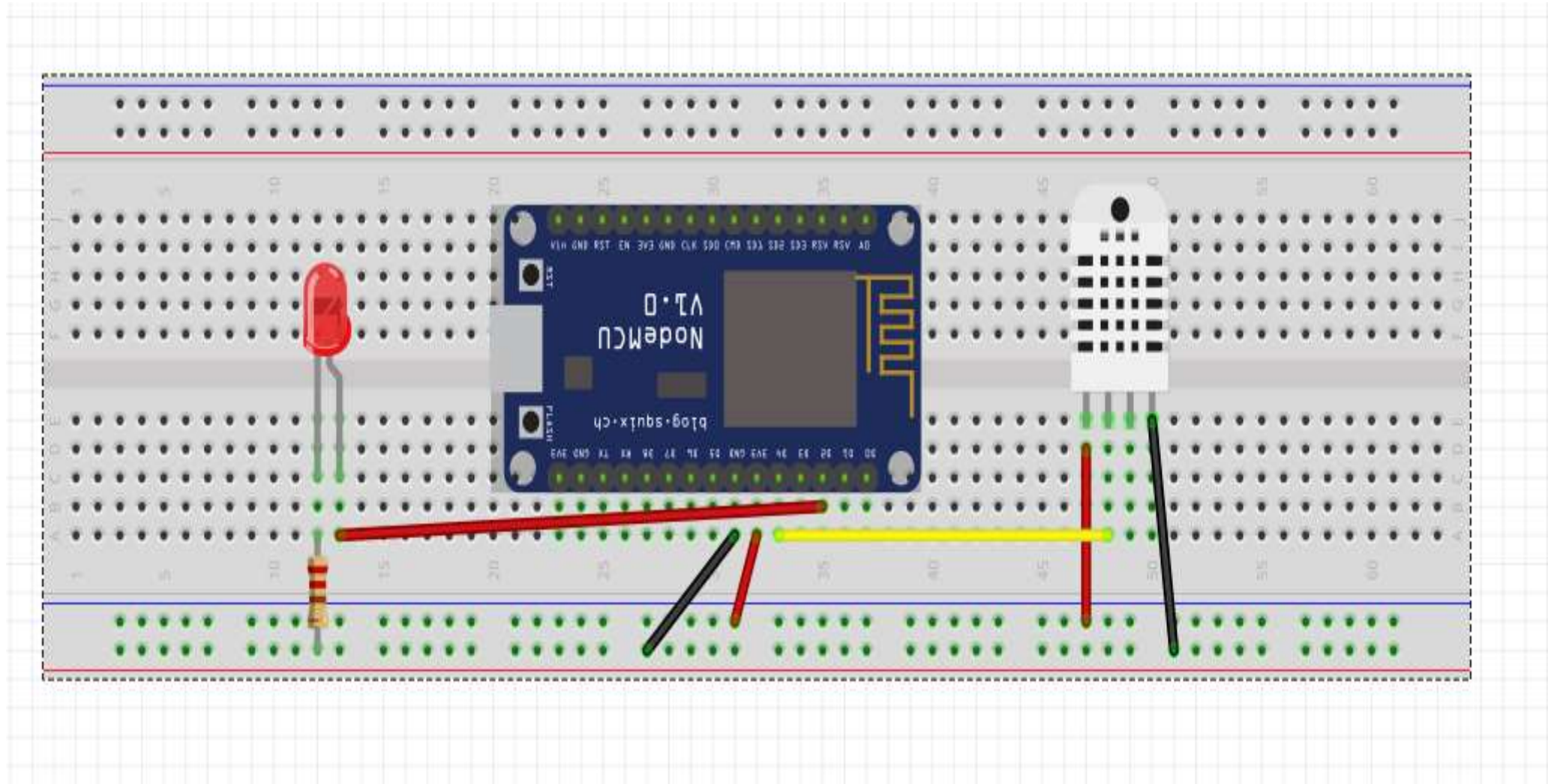
Node-RED interface showing the configuration for an MQTT in node. The configuration panel is titled "Edit mqtt in node".

Properties:

- Server: uniconverge mqtt
- Topic: sensor/humidity
- QoS: 0
- Output: auto-detect (string or buffer)
- Name: Name

The "Done" button is highlighted in red. The "Info" panel on the right shows the node's information, including its ID "1d26ad5a.7bb833" and type "mqtt in". The "Description" panel shows the node's function: "Connects to a MQTT broker and subscribes to messages from the specified topic." The "Outputs" panel shows the node's output format: "payload" (string | buffer), "topic" (string), "qos" (number), and "retain" (boolean).

Circuit diagram



Upload the code



The screenshot shows the Arduino IDE interface. The title bar reads "mqtt_temp_hum | Arduino 1.8.13". The menu bar includes "File", "Edit", "Sketch", "Tools", and "Help". The toolbar contains icons for opening, saving, and uploading. The "Upload" icon (a right-pointing arrow) is highlighted with a red square. Below the toolbar, the sketch name "mqtt_temp_hum" is displayed. The main text area contains the following code:

```
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include "DHT.h"

#define wifi_ssid "XXXXXXXXXX"
#define wifi_password "XXXXXXXXXX"

#define mqtt_server "test.uniconvergetech.in"
#define mqtt_user "guest"
#define mqtt_password "guest"

#define temperature_topic "sensor/temperature" //Topic temperature
#define humidity_topic "sensor/humidity" //Topic humidity

//Buffer to decode MQTT messages
char message_buff[100];

long lastMsg = 0;
```

Below the code, a status bar indicates "Done uploading." followed by two error messages in a black box:

```
Invalid library found in C:\Program Files (x86)\Arduino\libraries\async_mqtt_client: no headers files (.h) found in C:\Program Files (x86)\Arduino\libraries\async_mqtt_client
Invalid library found in C:\Program Files (x86)\Arduino\libraries\ESPAsyncTCP: no headers files (.h) found in C:\Program Files (x86)\Arduino\libraries\ESPAsyncTCP
```

The status bar at the bottom shows "16" on the left and "NodeMCU 1.0 (ESP-12E Module) on COM4" on the right.

Drag **debug node** from node palette

The screenshot shows the Node-RED web interface in a browser. The address bar displays the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface includes a top navigation bar with the Node-RED logo and a 'Deploy' button. Below this is a tabbed interface for managing flows, with 'Flow 4' currently selected. The left sidebar contains a 'filter nodes' search bar and two categories of nodes: 'common' and 'function'. In the 'common' category, the 'debug' node is highlighted with a red rectangular box. The main workspace shows a flow diagram with two nodes: 'sensor/temperature' and 'sensor/humidity'. The right sidebar displays the 'info' panel for the selected 'debug' node, showing its ID, type, and a detailed description of its function and outputs.

Node-RED

filter nodes

Flow 2 Flow 1 Flow 4

common

- inject
- debug**
- complete
- catch
- status
- link in
- link out
- comment

function

- function
- switch
- change

sensor/temperature

sensor/humidity

info

Information

Node	"1d26ad5a.7bb833"
Type	mqtt in

show more

Description

Node Help

Connects to a MQTT broker and subscribes to messages from the specified topic.

Outputs

payload	string buffer
a string unless detected as a binary buffer.	
topic	string
the MQTT topic, uses / as a hierarchy separator.	
qos	number
0, fire and forget - 1, at least once - 2, once and once only.	
retain	boolean
true indicates the message was retained	

Connect both mqtt in nodes to debug node & deploy

The screenshot shows the Node-RED web interface in a browser. The address bar displays the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface includes a top navigation bar with a "Deploy" button highlighted by a red box. On the left, a sidebar lists various nodes under "common" and "function" categories. The main workspace shows a flow named "Flow 4" with two MQTT sensor nodes, "sensor/temperature" and "sensor/humidity", both connected to a "msg.payload" node. This entire flow is enclosed in a red rectangular box. On the right, an "info" panel provides details about the selected flow, including its ID, name, and status.

Node-RED interface showing a flow with two MQTT sensor nodes (sensor/temperature and sensor/humidity) connected to a msg.payload node. The flow is highlighted with a red box. The interface also shows the Deploy button in the top right corner.

Flow 4

sensor/temperature

sensor/humidity

msg.payload

Deploy

info

Information

Flow	"aee7e802.c60bb8"
Name	Flow 4
Status	Enabled

Description

Both **mqtt in** nodes have connected to **broker** and data is receiving in output pane

The screenshot displays the Node-RED web interface in a browser. The address bar shows the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface includes a left sidebar with node categories (common and function), a central workspace with a flow diagram, and a right sidebar with a debug console.

Flow Diagram: The flow, titled "Flow 4", consists of two "mqtt in" nodes: "sensor/temperature" and "sensor/humidity". Both nodes show a green "connected" status. Their outputs are connected to a "msg.payload" node, which is a green box with a list icon.

Debug Console: The debug console on the right shows a list of messages. The messages are:

- 8/22/2020, 8:23:42 PM node: 413cc915.027c7
test : msg.payload : string[11]
"hello world"
- 8/22/2020, 8:29:43 PM node: 413cc915.027c7
test : msg.payload : string[17]
"testing data flow"
- 8/22/2020, 9:28:49 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.50"
- 8/22/2020, 9:28:49 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"86.00"

The last two messages, which represent the sensor data, are highlighted with a red box in the original image.

Drag **gauge node**(dashboard) from node palette into workspace

The screenshot shows the Node-RED web interface. On the left, the 'dashboard' category in the node palette is expanded, and the 'gauge' node is highlighted with a red rectangle. In the center workspace, a 'sensor/humidity' node is connected to 'Flow 4'. On the right, the 'debug' console displays a log of messages, including sensor data for humidity and temperature.

Node-RED interface showing the 'gauge' node (highlighted in red) in the dashboard node palette. The workspace contains a 'sensor/humidity' node connected to 'Flow 4'. The debug console on the right shows a log of messages, including sensor data for humidity and temperature.

Drag 2 gauges

The screenshot shows the Node-RED web interface. In the center workspace, a flow is visible with two sensor nodes: 'sensor/temperature' and 'sensor/humidity', both marked as 'connected'. These sensors are connected to a 'msg.payload' node. On the left sidebar, under the 'dashboard' category, various widgets are listed, including 'button', 'dropdown', 'switch', 'slider', 'numeric', 'text input', 'date picker', 'colour picker', 'form', 'text', 'gauge', and 'chart'. A red box highlights two 'gauge' widgets in the palette. A green arrow points from the text 'you need 2 guages' to this red box. On the right, the 'debug' console shows a series of log messages from 8/22/2020, 10:34:42 PM to 10:37:42 PM, displaying the values of the sensors: '88.00' for temperature and '29.50' for humidity.

you need 2 guages

debug

8/22/2020, 10:34:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:35:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.50"

8/22/2020, 10:35:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:36:43 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.50"

8/22/2020, 10:36:44 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:37:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:37:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

Double click on one of gauge

The screenshot shows the Node-RED web interface in a browser. The address bar displays the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface includes a left sidebar with a 'filter nodes' search bar and a 'dashboard' category expanded, showing various widgets like button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart. The main workspace shows a flow with two 'sensor' nodes (temperature and humidity) connected to a 'msg.payload' node. Two 'gauge' nodes are present in the workspace; the top one is highlighted with a red rectangular box, and the bottom one is outlined in orange. The top gauge node has a red triangle and a blue circle icon, while the bottom one has a red triangle and a blue circle icon.

Fill the details(Label-Temperature) & click on **done**

Node-RED interface showing the configuration for a gauge node. The configuration panel is titled "Edit gauge node".

Properties:

- Group: [Tab 1] Group 1
- Size: auto
- Type: Gauge
- Label: Temperature
- Value format: {{value}}
- Units: celcius
- Range: min 0, max 100
- Colour gradient: (Green, Yellow, Red bars)

The "Done" button is highlighted with a red box.

The background shows a flow with two sensors: "sensor/temperature" and "sensor/humidity", both connected.

The left sidebar shows a "dashboard" with various widgets: button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart.

The right sidebar shows a "debug" console with log messages:

```
8/22/2020, 10:40:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"87.00"

8/22/2020, 10:41:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:41:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"87.00"

8/22/2020, 10:42:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:42:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:43:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:43:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"87.00"
```

Double click on another gauge

The screenshot displays the Node-RED web interface in a browser window. The address bar shows the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface includes a left sidebar with a 'dashboard' menu and a list of nodes: button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart. The main workspace shows a flow with two 'sensor' nodes (temperature and humidity) connected to a 'msg.payload' node. The 'gauge' node is highlighted with a red box. The right sidebar shows the 'debug' console with a list of messages.

Flow 2 Flow 1 Flow 4

sensor/temperature connected

sensor/humidity connected

msg.payload

Temperature

gauge

debug

all nodes

8/22/2020, 10:42:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:43:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:43:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"87.00"

8/22/2020, 10:44:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:44:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:45:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:45:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

Fill the details (Label-Humidity) & click on **done**

The screenshot displays the Node-RED web interface. On the left, a sidebar contains various node types under the 'dashboard' category, including button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart. The main workspace shows a flow with two nodes: 'sensor/temperature' and 'sensor/humidity', both marked as 'connected'. The 'Edit gauge node' panel is open, showing configuration for a gauge node. The 'Group' is '[Tab 1] Group 1', 'Size' is 'auto', 'Type' is 'Gauge', 'Label' is 'Humidity', 'Value format' is '{{value}}', 'Units' is '%RH', and 'Range' is 'min 0 max 100'. The 'Colour gradient' section shows a green bar. The 'Done' button is highlighted in red. A green '5' is written above the 'Cancel' button. A green '2' is written next to the 'Label' field. A green '3' is written next to the 'Units' field. A green '4' is written next to the 'max' value in the range field. The debug console on the right shows log messages for sensor data.

debug

```
8/22/2020, 10:44:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:45:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:45:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:46:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:46:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"

8/22/2020, 10:47:42 PM node: a4969e3f.8b967
sensor/temperature : msg.payload : string[5]
"29.60"

8/22/2020, 10:47:42 PM node: a4969e3f.8b967
sensor/humidity : msg.payload : string[5]
"88.00"
```

Connect Nodes Mentioned In Slides & Deploy

The screenshot displays the Node-RED web interface in a browser. The address bar shows the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The top navigation bar includes a 'Deploy' button, which is highlighted with a red rectangle. The main workspace shows a flow diagram with the following nodes:

- sensor/temperature** (purple node, status: connected)
- sensor/humidity** (purple node, status: connected)
- msg.payload** (green node)
- Temperature** (blue node)
- Humidity** (blue node)

The flow diagram is enclosed in a red rectangle. The connections are as follows:

- sensor/temperature** connects to **msg.payload** and **Humidity**.
- sensor/humidity** connects to **msg.payload** and **Temperature**.
- msg.payload** connects to both **Temperature** and **Humidity**.

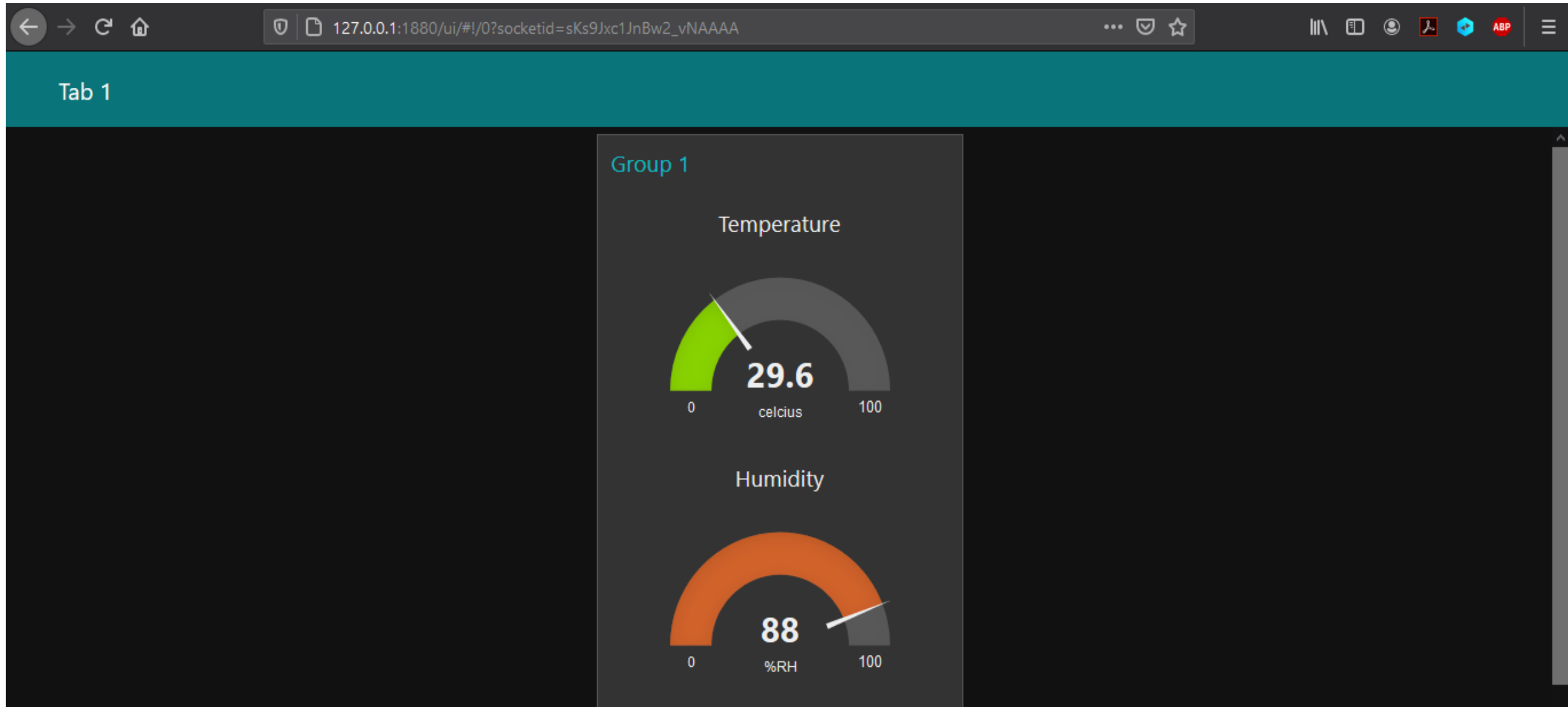
The right sidebar shows the 'debug' console with a list of log messages. The messages are as follows:

Time	Node ID	Log Message
8/22/2020, 10:46:42 PM	node: a4969e3f.8b967	sensor/humidity : msg.payload : string[5] "88.00"
8/22/2020, 10:47:42 PM	node: a4969e3f.8b967	sensor/temperature : msg.payload : string[5] "29.60"
8/22/2020, 10:47:42 PM	node: a4969e3f.8b967	sensor/humidity : msg.payload : string[5] "88.00"
8/22/2020, 10:48:42 PM	node: a4969e3f.8b967	sensor/temperature : msg.payload : string[5] "29.50"
8/22/2020, 10:48:42 PM	node: a4969e3f.8b967	sensor/humidity : msg.payload : string[5] "88.00"
8/22/2020, 10:49:42 PM	node: a4969e3f.8b967	sensor/temperature : msg.payload : string[5] "29.60"
8/22/2020, 10:49:42 PM	node: a4969e3f.8b967	sensor/humidity : msg.payload : string[5] "88.00"

Click on dashboard tab in output pane

The screenshot displays the Node-RED web interface in a browser. The address bar shows the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The main workspace shows a flow with two sensor nodes: `sensor/temperature` and `sensor/humidity`, both marked as connected. These sensors feed into a `msg.payload` node, which then connects to `Temperature` and `Humidity` display nodes. On the left, a sidebar lists various dashboard widgets like button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart. On the right, the 'dashboard' output pane is active, showing tabs for Layout, Site, and Theme. A green number '2' is placed over the Theme tab. A red box highlights the 'dashboard' tab icon at the top of the output pane, and another red box highlights the 'link' icon in the 'Tabs & Links' section. Below the 'Tabs & Links' section, a message states: 'There are 6 widgets not in a group. Click here to create the missing groups'.

Dashboard



Drag a switch node(dashboard in node palette) into workspace

The screenshot displays the Node-RED web interface in a browser. The address bar shows the URL `127.0.0.1:1880/#flow/aee7e802.c60bb8`. The interface is divided into three main sections:

- Node Palette (Left):** A sidebar containing various nodes. The **dashboard** category is expanded, and the **switch** node is highlighted with a red rectangle. Other nodes visible include button, dropdown, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart.
- Workspace (Center):** The main area where a flow is being built. It contains two sensor nodes: **sensor/temperature** and **sensor/humidity**, both marked as "connected". These sensors are connected to a **msg.payload** node, which then branches out to two output nodes: **Temperature** and **Humidity**.
- Dashboard Sidebar (Right):** A sidebar for managing the dashboard. It includes tabs for **Layout**, **Site**, and **Theme**. Below these is the **Tabs & Links** section, showing a tree structure with **Tab 1** and **Group 1**. A message at the bottom states: "There are 6 widgets not in a group. Click here to create the missing groups".

Double click on **switch node**

The screenshot displays the Node-RED web interface in a browser window. The address bar shows the URL `127.0.0.1:1880/#flow/ae7e802.c60bb8`. The interface includes a left sidebar with a 'filter nodes' search bar and a 'dashboard' category expanded, showing various widget options like button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart. The main workspace shows a flow named 'Flow 4' with the following components: two sensor nodes ('sensor/temperature' and 'sensor/humidity', both marked 'connected'), a 'msg.payload' node, and two output nodes ('Temperature' and 'Humidity'). Connections show both sensors feeding into the 'msg.payload' node, which then feeds into the 'Temperature' output node. The 'sensor/humidity' node also has a direct connection to the 'Humidity' output node. A 'switch' node is located in the lower-left area of the workspace, highlighted with a red rectangular box. The right sidebar contains a 'dashboard' panel with tabs for 'Layout', 'Site', and 'Theme', and a 'Tabs & Links' section showing a tree structure with 'Tab 1' and 'Group 1'.

Fill details

The screenshot displays the Node-RED web interface. On the left, the 'dashboard' palette lists various widgets: button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart. The main workspace shows a flow with two sensor nodes ('sensor/temperature' and 'sensor/humidity') and a switch node. The 'Edit switch node' configuration panel is open on the right, showing the following settings:

- Delete** button
- Cancel** button
- Done** button
- Properties** section:
 - Tooltip**: optional tooltip
 - Icon**: Default
 - On Icon**: [input field] Colour [input field]
 - Off Icon**: [input field] Colour [input field]
 - If msg arrives on input, pass through to output**: ☒
 - When clicked, send**:
 - On Payload**: a_z ON ➡
 - Off Payload**: a_z OFF ➡
 - Topic**: [input field]
 - Name**: [input field]
- Enabled** checkbox

On the far right, the 'dashboard' configuration panel shows the 'Layout' tab with 'Tab 1' containing a 'Group 1' with widgets for 'Temperature', 'Humidity', and 'switch'.

Select same group which is selected in **gauge nodes** & click on **done**

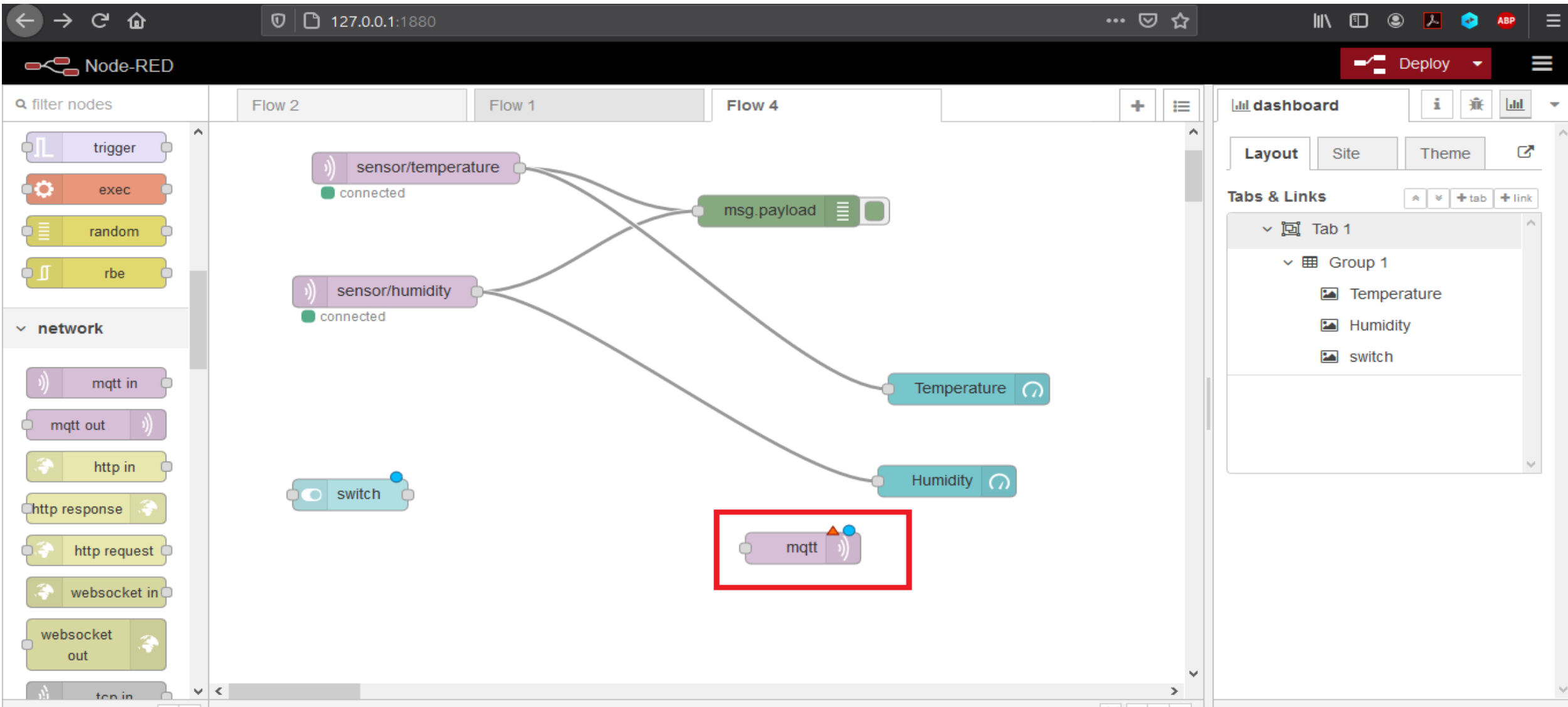
The screenshot displays the Node-RED web interface in a browser window. The address bar shows the URL `127.0.0.1:1880`. The interface is divided into several sections:

- Left Sidebar (Nodes):** A list of nodes under the 'dashboard' category, including button, dropdown, switch, slider, numeric, text input, date picker, colour picker, form, text, gauge, and chart.
- Flow Canvas:** The central workspace showing a flow with two sensor nodes (temperature and humidity) connected to a switch node.
- Right Panel (Edit switch node):** A configuration dialog for the selected switch node. It includes fields for Group, Size, Label, Tooltip, Icon, and On/Off icons. The 'Group' field is set to '[Tab 1] Group 1' and is highlighted with a red box. The 'Done' button at the top right of the dialog is also highlighted with a red box.
- Far Right Panel (Dashboard):** A preview of the dashboard layout, showing a tab named 'Tab 1' containing a group named 'Group 1' with three items: Temperature, Humidity, and switch.

Drag **mqtt out** node (from node palette) into workspace

The screenshot displays the Node-RED web interface in a browser window. The address bar shows the URL `127.0.0.1:1880`. The interface includes a top navigation bar with a 'Deploy' button and a sidebar on the left containing a 'filter nodes' search bar and a list of nodes categorized under 'network'. The 'mqtt out' node is highlighted with a red rectangular box in the network category. The main workspace, titled 'Flow 4', contains a workflow with two 'sensor' nodes ('sensor/temperature' and 'sensor/humidity'), a 'msg.payload' node, and two output nodes ('Temperature' and 'Humidity'). The 'mqtt in' node is also visible in the network category. The right sidebar shows a 'dashboard' panel with 'Layout', 'Site', and 'Theme' tabs, and a 'Tabs & Links' section with a 'Tab 1' group containing 'Temperature', 'Humidity', and 'switch' links.

Double click on **mqtt out** node



Fill the details & write topic **led/switch1**

The screenshot shows the Node-RED web interface in a browser window. The address bar displays '127.0.0.1:1880'. The main workspace shows a flow with two sensors (temperature and humidity) connected to a switch node. The 'network' sidebar on the left contains various nodes, including 'mqtt in' and 'mqtt out'. The 'Edit mqtt out node' panel is open on the right, showing the configuration for an MQTT output node. The 'Server' field is set to 'uniconverge mqtt' (marked with a green '1'), and the 'Topic' field is set to 'led/switch1' (marked with a green '2'). The 'Done' button is highlighted with a red box and a green '3'. The 'Properties' section includes fields for QoS, Retain, and Name. A tip message at the bottom of the panel reads: 'Tip: Leave topic, qos or retain blank if you want to set them via msg properties.' The right sidebar shows the 'dashboard' tab with a 'Layout' section and a 'Tabs & Links' section containing a 'Tab 1' with a 'Group 1' containing 'Temperature', 'Humidity', and 'switch'.

Node-RED interface showing the configuration of an MQTT output node.

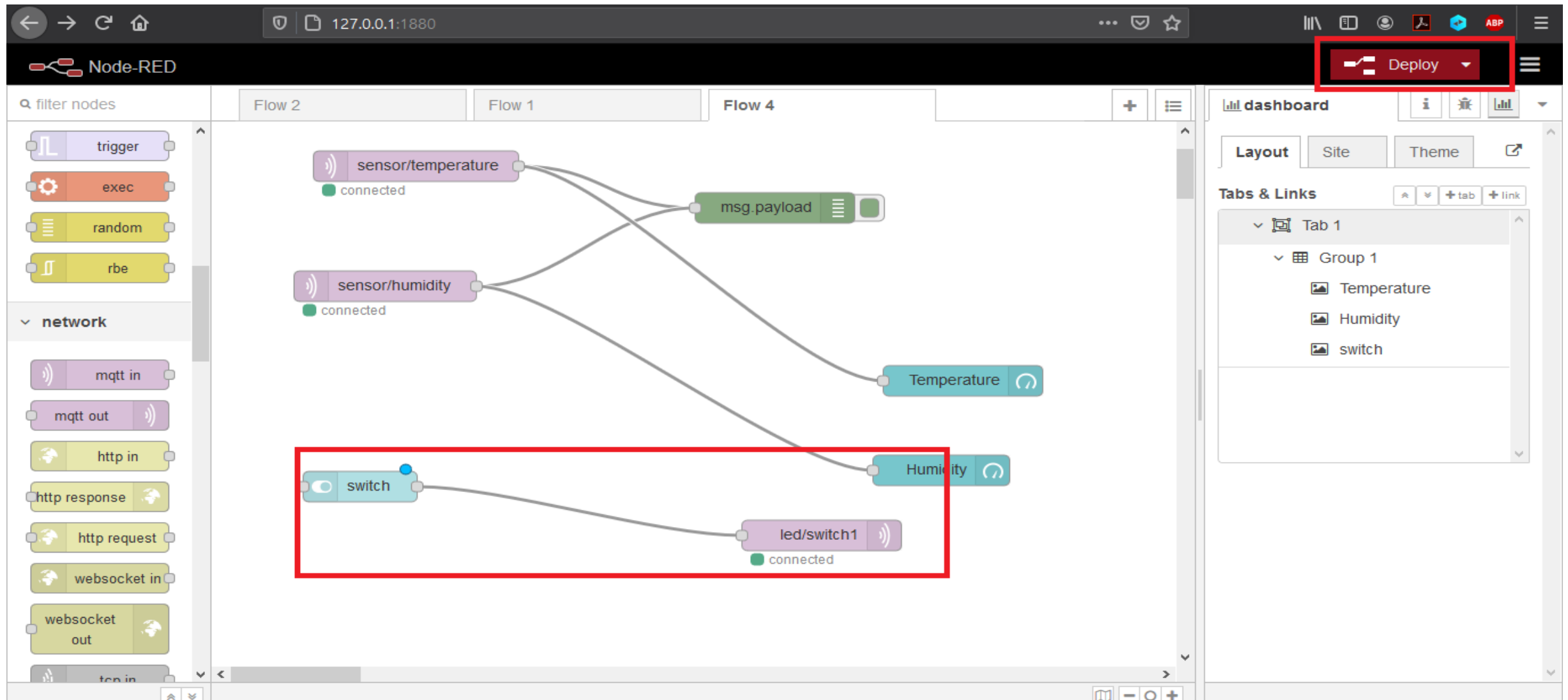
The configuration panel displays the following details:

- Server: uniconverge mqtt (1)
- Topic: led/switch1 (2)
- QoS: (blank)
- Retain: (blank)
- Name: Name

The 'Done' button is highlighted (3).

Tip: Leave topic, qos or retain blank if you want to set them via msg properties.

Connect switch node and mqtt out node & deploy



Switch is showing on dashboard

